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Catching mRNA's Hiddens Marks: A Dual-Path Network by Contrastive Learning for N4-acetylcytidine Prediction

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Abstract. N4-acetylcytidine (ac4C) is a crucial RNA modification associated with mRNA stability and translational efficiency. Accurate identification of ac4C sites is essential for understanding their regulatory functions. However, experimental detection remains expensive and labor-intensive. At the same time, existing computational models suffer from limited generalization and insufficient feature discrimination, especially in distinguishing subtle nucleotide patterns. In this work, we propose a deep learning model named SNN-ac4C, which is based on a contrastive learning-based neural network. The model integrates a dual-path structure that combines BiLSTM and Multi-Head Self-Attention (MHSA) for capturing long-range dependencies and global context, while using CNN to extract local biological sequence features. The contrastive learning module further enhances the discriminative ability of ac4C and Non-ac4C sites by increasing the separation between positive and negative samples. Experiments on the test set confirm the effectiveness of SNN-ac4C, which achieves an accuracy (ACC) of 84.60% and a Matthews Correlation Coefficient (MCC) of 0.6934. Compared with NBCR-ac4C, the current state-of-the-art model, SNN-ac4C improves ACC and MCC by 1.09% and 0.0228, respectively. The source code and relevant supplementary are publicly available at https://github.com/2103374200/SNN.

1 Introduction

RNA modifications play a crucial role in regulating gene expression and maintaining normal cellular function. The MODOMICS database [4] reports over 170 distinct types of RNA modifications. Several well-known mRNA modifications include N6-methyladenosine (m6A) [8], guanosine methylation (Gm) [36], adenosine methylation (Am)[6], 5-formylcytidine (f5C) [16], N1-methyladenosine (m1A) [35], cytidine methylation (Cm) [33], 5-methylcytidine (m5C) [14], and N4-acetylcytidine (ac4C) [17]. The ac4C, formed by the acetylation at the N4 position of cytidine, emerges as a key epitranscriptomic marker. Its modification process depends on specific enzymes. As a unique chemical marker, ac4C improves mRNA stability and alters its secondary structure. It also modulates molecular interactions and participates in key biological processes, such as RNA-protein binding [31].

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Recent biomedical studies reveal that ac4C modifications are associated with multiple human diseases [17]. These include epithelial ovarian cancer [41], gestational diabetes mellitus [21], pulmonary fibrosis [19], interstitial cystitis [26], chronic kidney disease [25], AIDS [5], cirrhosis [3], and breast cancer [22]. How to accurately identify ac4C modification sites proves to be essential, due to this supports further investigation in molecular biology and guides the development of targeted therapies. Although the rapid increase in sequencing data creates opportunities for identification, it introduces challenges to site prediction because of the subtle differences between modified and unmodified sequences.

Currently, researchers propose a variety of approaches, including experimental techniques, traditional machine learning models, and modern deep learning frameworks [28, 10, 24]. However, most of the existing methods face two major limitations. First of all, they often fail to simultaneously capture local features and global dependencies, both of which are important for discovering ac4C sites. Then, weak supervised models often generalize poorly to independent datasets [20]. These limitations highlight the need for improved feature extraction methods to obtain a discriminative feature representation for mRNA sequences.

To solve these problems, we propose SNN-ac4C, a novel model that adopts a refined deep learning framework. Specifically, it leverages a dual-path contrastive learning approach [38]. One path consists of a convolutional neural network (CNN) that captures local patterns to preserve essential biological signals [37] in mRNA sequences. The other path combines bidirectional long short-term memory (BiLSTM) and multi-head self-attention (MHSA). This branch learns contextual relationships and handles global dependencies. To further improve generalization, SNN-ac4C incorporates contrastive learning. This training strategy enhances representation quality by bringing positive pairs (belong to the same class) closer and pushing apart dissimilar ones (belong to the different classes) [18].

We evaluate the performance of SNN-ac4C on a widely used benchmark dataset [30]. Comparison experiments with other state-of-the-art methods, such as NBCR-ac4C, show that our model performs better. Both validation and independent testing sets produce consistently superior prediction results. We conduct ablation experiments to assess the impact of each module. The results confirm the effectiveness of both the contrastive learning strategy and the MHSA mechanism. Furthermore, robustness tests on the dataset derived from PACES [42] demonstrate the model's ability to maintain

stable predictions under varying conditions and sequence lengths. SNN-ac4C achieves an accuracy (ACC) of 84.60% and a Matthews correlation coefficient (MCC) of 0.6934. These values exceed those of NBCR-ac4C by 1.09% in ACC and 0.0228 in MCC. The results clarify the generalization and practical value of the given SNN-ac4C.

In summary, our work presents the following key contributions:

- We present SNN-ac4C, a deep learning framework that effectively integrates both local features and global dependencies for accurate ac4C site identification in mRNA sequence.
- We construct a dual-path architecture that combines CNN, BiLSTM, Multi-Head Self-Attention, and contrastive learning to enhance feature representation and model generalization.
- We conduct comprehensive evaluations showing that SNN-ac4C achieves state-of-the-art performance, reaching 84.60% accuracy and 0.6934 MCC on benchmark dataset.

2 Related Works

Existing models focus on specific neural network designs for ac4C site prediction. NBCR-ac4C [10] uses Transformer-based embeddings with ResNet18 to extract contextual and hierarchical features. Stacking-ac4C [24] combines multiple encoding strategies with ensemble classification to enhance robustness. DPNN-ac4C [39] adopts deeper attention networks to capture nucleotide dependencies. TransC-ac4C [23] also merges multiple encodings for both local and long-range dependencies. While these models improve accuracy, most do not explicitly separate local and global feature learning [20], potentially missing structural differences in mRNA. We propose a dual-path design that facilitates diverse pattern extraction.

Effective representation learning is essential for ac4C site prediction, especially when labeled data is scarce. Prior machine learning methods such as PACES [42] and ac4C-AFL [28] rely on predefined features such as dinucleotide profiles and frequency vectors. These features limit the expressive power of models and require domain-specific knowledge. Deep models like NBCR-ac4C [10] improve this deficiency by learning features from sequences directly. But they usually optimize only for the main classification task. They do not leverage auxiliary signals that can refine embeddings. In contrast, we introduce contrastive learning for embedding enhancement to guide encoder to generating more discriminative feature representations.

Generalization is also a critical issue for ac4C predictors, especially when applied to new cell lines or sequence datasets. The general models such as ac4C-AFL and DPNN-ac4C report high accuracy on held-out data sampled from the same distribution as training data [28, 39]. However, their performance frequently drops on unseen or heterogeneous samples. Such gaps reveal that these models may overfit to dataset-specific biases [12]. As a result, it is urgent to use both random and independent test sets to assess model robustness. Our approach combines global-local learning with contrastive learning-based regularization, which benefits the generation of more stable performance across diverse experiment datasets.

3 Methodology

3.1 Dataset

To evaluate the performance of SNN-ac4C, we adopt the refined benchmark dataset introduced by Su et al. [30]. It originates from the real experiments, which comprises 4250 candidate ac4C peaks identified through acRIP-seq [1]. Initially, the cytosines nearest to the ac4C peaks serve as the modification sites. We extract 100 nucleotides from both upstream and downstream of these sites to construct positive samples. For negative samples, we randomly select regions located outside the peaks. Each negative sequence spans 201 nucleotides and places a cytosine at the center. This sampling strategy simulates the characteristics of positive samples and maintains dataset balance. We also apply the CD-HIT tool [7] with a similarity threshold of 0.8 to remove redundant sequences and improve predictive performance. In addition, we randomly choose the same number of negative samples as positive ones to ensure class balance. Finally, we split this dataset [30] into a training set and an independent test set at a 4:1 ratio. The training set contains 2206 positive and 2206 negative samples. The test set has 552 positive and 552 negative samples.

3.2 Feature Extraction

3.2.1 Embedding-based Features

In RNA sequence analysis, nucleotide sequences consist of discrete characters (A, U, G, C). Deep learning models require these characters to be converted into numerical inputs. Researchers typically apply an integer encoding method to mapping each nucleotide into a unique integer. In this way, a vocabulary is constructed by extracting all possible nucleotide symbols from the RNA dataset. For every sequence, we define the vocabulary as {'A': 3, 'C': 2, 'G': 1, 'U': 4}. Each nucleotide in the input sequence is replaced with its corresponding integer. For instance, given the sequence S = AGGUCCUGA, the integer encoding produces: $[3, 1, 1, 4, 2, 2, 4, 1, 3]$. Compared with one-hot encoding, integer encoding provides a more compact storage format and less computational cost. Additionally, this definition integrates smoothly with embedding layers, which allows the model to optimize symbol representations directly.

In this work, we employ the embedding layer from PyTorch (`nn.Embedding`) as the main module for feature extraction (see Fig. 1 A). The `nn.Embedding` layer acts as a learnable lookup table. It maps discrete indices to high-dimensional continuous vectors that produce a specific representation for each input symbol. The integer-encoded sequence is passed through the embedding layer. This process yields a dense and continuous vector representation. The embedding layer assigns each integer a d-dimensional vector. As a result, the entire sequence becomes an $L \times d$ embedding matrix, where L is the sequence length and d is the embedding dimension. In our settings, we choose $d = 800$ to balance model capacity and computational efficiency. The embedding matrix is updated using gradient descent during training. This enables the embedding vectors to capture semantic information relevant to the task.

3.2.2 SCPseDNC-based Features

Series Correlation Pseudo Dinucleotide Composition (SCPseDNC) is a feature extraction method widely applied in DNA and RNA sequence analysis tasks, such as functional prediction, classification, and clustering [40]. It extends the traditional dinucleotide composition (DNC) approach. SCPseDNC fuses the relative frequencies of dinucleotides with the physicochemical correlations among them. This combination provides a richer feature representation for biological sequence analysis. SCPseDNC encoding is defined as follows:

$$V = [d_1, d_2, ..., d_{16}, d_{16+\lambda}, d_{16+\lambda+1}, ..., d_{16+\lambda\Lambda}]^T \quad (1)$$

where:

$$d_u = \begin{cases} \frac{f_u}{\sum_{i=1}^{16} f_i + w\sum_{j=1}^{\lambda\Lambda}\theta_j}, & (1 \le u \le 16) \\ \frac{w\theta_{u-16}}{\sum_{i=1}^{16} f_i + w\sum_{j=1}^{\lambda\Lambda}\theta_j}, & (17 \le u \le 16+\lambda\Lambda) \end{cases} \quad (2)$$

where f_u ($u = 1, 2, ..., 16$) is the normalized occurrence frequency of the dinucleotide in the sequence, w is the weight factor ranged from 0 to 1, Λ is the number of physicochemical indices. θ_j ($j = 1, 2, ..., \lambda$) is the j-tier correlation factor that is defined as:

$$\begin{cases} \theta_1 = \frac{1}{L-3}\sum_{i=1}^{L-3} J_{i,i+1}^1 \\ \theta_2 = \frac{1}{L-3}\sum_{i=1}^{L-3} J_{i,i+1}^2 \\ \cdots \\ \theta_\Lambda = \frac{1}{L-3}\sum_{i=1}^{L-3} J_{i,i+1}^\Lambda \qquad (\lambda < L-2) \\ \cdots \\ \theta_{\lambda\Lambda-1} = \frac{1}{L-\lambda-2}\sum_{i=1}^{L-\lambda-2} J_{i,i+\lambda}^{\Lambda-1} \\ \theta_{\lambda\Lambda} = \frac{1}{L-\lambda-2}\sum_{i=1}^{L-\lambda-2} J_{i,i+\lambda}^{\Lambda} \end{cases} \quad (3)$$

λ represents the highest counted rank (or tier) of the correlation along the nucleotide sequence, and the correlation function is:

$$J_{i,i+m}^u = P_u(R_iR_{i+1}) \cdot P_u(R_{i+m}R_{i+m+1}) \quad (4)$$

where u is the number of physicochemical indices. In this study, six indices ('Rise', 'Roll', 'Shift', 'Slide', 'Tilt', 'Twist') are used. $P_u(R_iR_{i+1})$ is the numerical value of the u-th physicochemical index of dinucleotide R_iR_{i+1} at position i and $P_u(R_jR_{j+1})$ represents the corresponding value of dinucleotide R_jR_{j+1} at position j.

SCPseDNC captures local composition and global sequence dependency at the same time. This improves the accuracy of sequence analysis. Its ability to include various physicochemical descriptors also enhances the biological interpretability of the model [40]. Therefore, SCPseDNC remains suitable for feature extraction in both RNA and DNA sequence tasks. In this study, we adopt SCPseDNC for manual feature extraction (see Fig. 1 C).

3.3 SNN-ac4C

3.3.1 BiLSTM Module

Bidirectional Long Short-Term Memory (BiLSTM) is an improved version of the Recurrent Neural Network (RNN). It integrates LSTM units with a bidirectional architecture. This structure enables the model to learn information from both forward and backward directions [2]. In our framework, the embedding matrix obtained in the Section 3.2.1 serves as the input x_i to the BiLSTM module. As shown in Fig. 1 F, the BiLSTM layer processes this embedded representation to extract contextual features more effectively.

BiLSTM is based on the LSTM structure. It uses two separate LSTM networks. One processes the input sequence from left to right. The other reads the sequence in reverse. This design allows the model to incorporate both past and future context into the representation of each element. As a result, the output O_i at each position carries richer semantic information. The bidirectional setting is particularly useful in tasks that rely on a complete understanding of sequence context. Meanwhile, the LSTM gating mechanism continues to mitigate issues related to long-range dependencies. Experimental results in Section 4.2 demonstrate that BiLSTM, which reflects deep sequence-level features, improves the modeling of both global and local sequence patterns. This enhancement contributes to more accurate identification of mRNA modification sites.

3.3.2 Multi-Head Self-Attention Module

The output vectors from the BiLSTM layer serve as the input to the multi-head self-attention (MHSA) module, as shown in Fig. 1 G. These vectors provide contextualized representations that preserve both forward and backward dependencies. At the same time, MHSA further strengthens the ability to capture complex dependencies, via calculating the correlation between each element in these input vectors and every other element [34].

In this study, the number of attention heads is set to $h = 8$. We apply the MHSA to compute attention in parallel across multiple subspaces. Each head emphasizes different parts of the sequence and learns various features across different subspaces. Given an input vector, this module first maps it into three vector spaces, Query (Q), Key (K), and Value (V).

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V \quad (5)$$

The W_Q, W_K, and W_V represent learnable weight matrices. The attention scores result from the dot product between Query and Key, followed by scaling:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (6)$$

where the scaling factor $\sqrt{d_k}$ reduces the magnitude of dot products. This helps avoid vanishing gradients during training.

The MHSA further splits the input into h different subspaces. Each subspace computes attention independently. The results are then concatenated. Specifically, for each head i, the input sequence is projected into distinct query, key, and value spaces:

$$Q_i = XW_Q^i, \quad K_i = XW_K^i, \quad V_i = XW_V^i \quad (7)$$

where the attention output for the i-th head is given by: $\text{head}_i = \text{Attention}(Q_i, K_i, V_i)$. The outputs are combined after calculating the attention for all heads. A linear transformation is then applied to produce the final output:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \text{head}_2, \ldots, \text{head}_h)W_O \quad (8)$$

3.3.3 CNN Module

Section 3.3.1 and Section 3.3.2 take the embedding-based features to model sequential dependencies and contextual information, while we employ the SCPseDNC-based features (Section 3.2.2) in this section by convolutional neural network (CNN) to extract local compositional patterns and physicochemical correlations. This dual-feature design allows the model to leverage both semantic representations and biologically structural information.

In general, CNN is widely applied in RNA secondary structure prediction, modification site identification (e.g., m6A methylation sites), and splicing site prediction [10]. They are also used

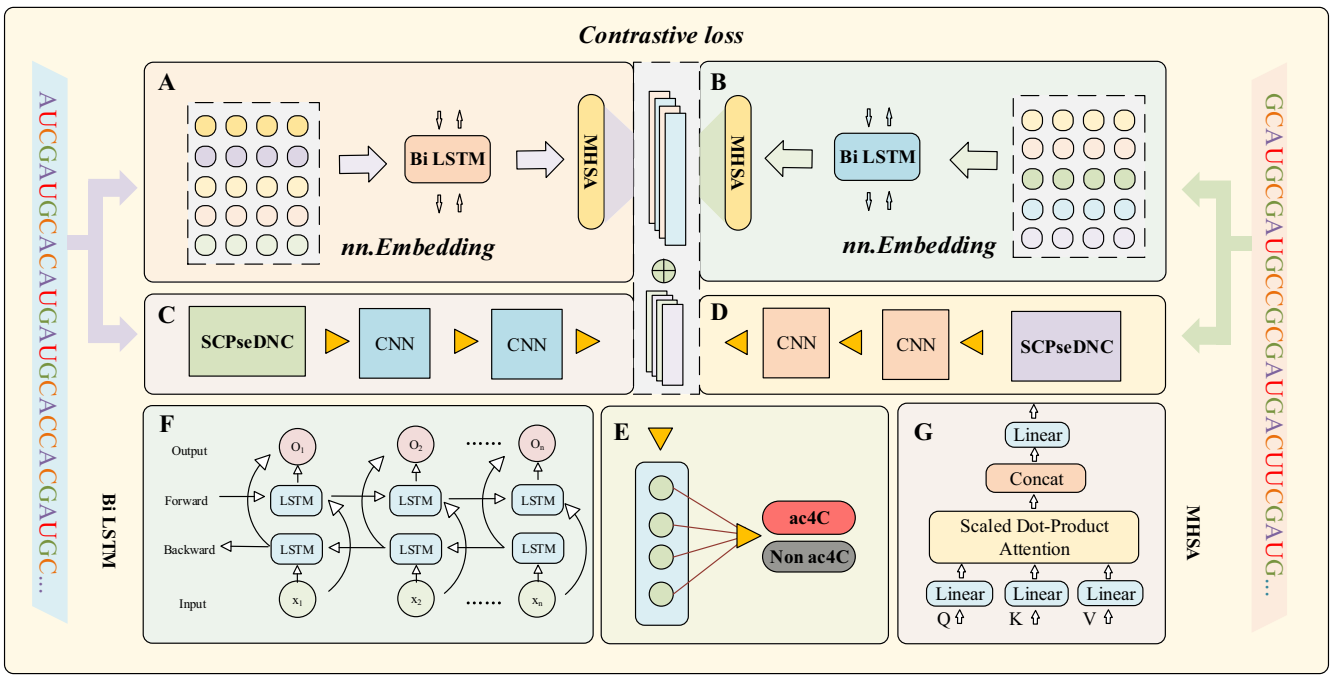


Figure 1. Overview of SNN-ac4C.

in modeling RNA secondary structures and RNA–protein interactions [29, 27]. Its local feature extraction ability makes it available to learn features directly from RNA sequences and structures. We design a two-layer CNN module to receive the traditional handcrafted SCPseDNC-based features. It enhances the performance of SNN-ac4C in ac4C site recognition by learning both local compositional characteristics and global sequence dependencies (see Fig. 1 C).

3.3.4 *Contrastive Learning Module*

The embedding-based feature vectors processed by BiLSTM and attention, as well as the SCPseDNC-based feature vectors transformed by CNN, are fed into our designed contrastive learning model. It aims to maximize the similarity between positive sample pairs and minimize the similarity between negative sample pairs, which helps learn feature representations and enhances indentification ability [37].

We design a Dual-Path Feature Extraction Module (Path 1 & Path 2). This module handles mRNA fragments from different sources. It forms positive and negative sample pairs for contrastive learning. For positive pairs, which consist of augmented versions of the same mRNA fragment, the extracted features remain similar. For negative pairs, composed of different mRNA fragments, the features should differ. A contrastive loss function is adopted during training. We replace the conventional Euclidean distance with a negative cosine similarity as the distance metric. This metric is defined as:

$$D(v_1, v_2) = 1 - \frac{v_1 \cdot v_2}{\|v_1\|\|v_2\|} \tag{9}$$

where v_1 and v_2 represent the feature vectors extracted by Path 1 and Path 2, respectively. We then, by applying this distance, define the final contrastive loss as

$$L_1(v_1, v_2, Y) = (1-Y)\cdot\frac{1}{2}\,[D(v_1, v_2)]^2 + Y\cdot\frac{1}{2}\max(0, m - D(v_1, v_2))^2 \tag{10}$$

In this formula, m denotes a hyperparameter (margin) that specifies the minimum separation between negative sample pairs. The label Y indicates the pair type, where $Y = 0$ corresponds to positive pairs and $Y = 1$ to negative pairs.

By minimizing the contrastive loss, the model aligns representations of positive samples closely. At the same time, it separates negative samples in the feature space. This learning process improves the accuracy of ac4C site prediction and enhances generalization ability.

3.3.5 *Fully Connected Layer Module*

The features produced from the contrastive learning module are passed into a fully connected (FC) layer in Fig. 1 E. The FC layer combines feature vectors to aggregate complex relationships among input sequences. Due to a high-dimensional feature vector that may increase the risk of overfitting, we employ Dropout [13] layer to randomly deactivate a portion of neurons during training. The dropout rate is set to 0.5 In the experiment. This setting introduces stochasticity to enlarge robustness and control model complexity. The Softmax activation function [12] is applied to turn raw output values into a normalized probability distribution. For the recognition of the ac4C site, a binary classification task, softmax assigns a probability for each input sequence to predict they are "ac4C" or "Non-ac4C".

3.4 *Training and Testing*

We separate the sample in the training set, including 2206 positive and 2206 negative samples, into training and validation sets by a random split strategy of 9:1, in which training set trains the SNN-ac4C, and validation set evaluates performance. The Adam optimizer is used with an initial learning rate 1×10^{-4} [12]. The batch size is set to 32. We adopt a joint optimization strategy that combines binary cross-entropy loss L_2 with contrastive loss L_1 during training.

For the margin hyperparameter m in the contrastive loss function (Equation 10), we conduct a grid search on the validation set to determine the optimal value. This process ensures an optimal balance between effective class separation and stable model training. The grid search evaluates different margin values to find the one that maximizes the model's performance while maintaining training stability.

For each sequence in a paired sequence, we compute the binary cross-entropy loss as follows:

$$L_2(v_1, Y_1) = -\left[Y_1 \log(y(v_1)) + (1 - Y_1) \log(1 - y(v_1))\right] \quad (11)$$

$$L_2(v_2, Y_2) = -\left[Y_2 \log(y(v_2)) + (1 - Y_2) \log(1 - y(v_2))\right] \quad (12)$$

where Y_1 and Y_2 represent the labels for sequences S_1 and S_2, respectively. $y(v_i)$ denotes the predicted probability that sequence S_i belongs to the positive class. The overall loss function is the sum of the contrastive loss and the binary cross-entropy losses:

$$L(v_1, v_2, Y, Y_1, Y_2) = L_1(v_1, v_2, Y) + L_2(v_1, Y_1) + L_2(v_2, Y_2) \quad (13)$$

We conduct all steps (containing model training, etc.) on a server with Ubuntu 20.04.6 LTS an Intel(R) Xeon(R) Platinum 8336C processor and an NVIDIA GeForce RTX 4090 graphics card.

3.5 Metrics

We evaluate performance via five key metrics [9]: Sensitivity (SEN), Specificity (SPE), Accuracy (ACC), Matthews Correlation Coefficient (MCC), and Area Under the Receiver Operating Characteristic Curve (AUROC). Sensitivity reflects the model's ability to correctly identify true positives. Specificity describes its ability to correctly exclude negative cases. Accuracy indicates the proportion of correctly predicted samples. It serves as an intuitive indicator of overall effectiveness. MCC provides a more balanced assessment. It considers true positives, true negatives, false positives, and false negatives simultaneously. AUROC evaluates classification performance by examining the trade-off between true positive and false positive rates.

4 Experiments

4.1 Comparison with General Deep Learning Models

Table 1. Prediction results on validation set, compared with deep learning models.

Model	SEN(%)	SPE(%)	ACC(%)	MCC	AUROC
CNN	68.78	81.00	74.89	0.5015	0.8457
LSTM	**88.24**	74.21	81.22	0.6307	0.8910
LSTM+Att	82.81	80.54	81.67	0.6336	0.8857
GRU	85.07	77.83	81.45	0.6306	0.8863
GRU+Att	82.35	81.45	81.90	0.6380	0.8748
SNN-ac4C	83.26	**82.35**	**82.81**	**0.6561**	**0.8911**

We compare five widely-used deep learning models—CNN, LSTM, LSTM+Att, GRU, and GRU+Att—with SNN-ac4C. As shown in Table 1, SNN-ac4C achieves a SEN of 83.26%, a SPE of 82.35%, an ACC of 82.81%, and an MCC of 0.6561 on the validation set. The AUROC reaches 0.8911. The plain LSTM model reports a higher SEN of 88.24%. However, its SPE drops to 74.21%, resulting in a less balanced performance. In contrast, SNN-ac4C demonstrates a better trade-off between SEN and SPE. This balance is also reflected in its higher ACC and MCC.

Table 2. Prediction results on the test set, compared with deep learning models.

Model	SEN(%)	SPE(%)	ACC(%)	MCC	AUROC
CNN	83.88	67.21	75.54	0.5181	0.8328
LSTM	80.98	**83.70**	82.34	0.6470	0.9004
LSTM+Att	80.70	83.15	81.61	0.6325	0.8911
GRU	80.80	82.61	81.70	0.6342	0.8830
GRU+Att	81.34	81.16	81.25	0.6250	0.8690
SNN-ac4C	**87.68**	81.52	**84.60**	**0.6934**	**0.9022**

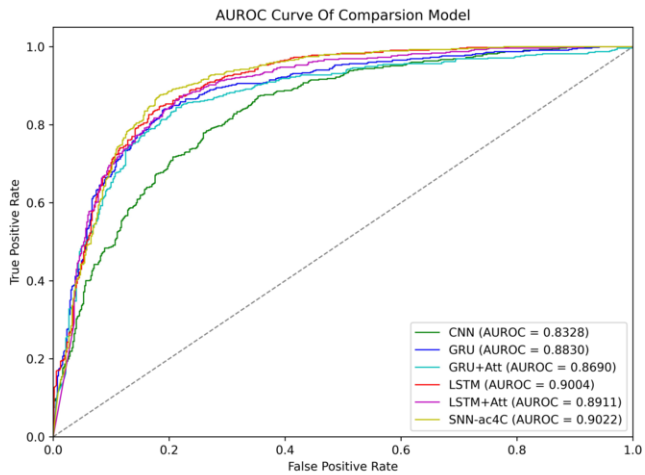


Figure 2. AUROC curves on test set, compared with deep learning models.

Results on independent test set (Table 2 and Fig. 2) further confirm the robustness of SNN-ac4C. The model gets a SEN of 87.68%, an ACC of 84.60%, and an MCC of 0.6934. It maintains a competitive SPE of 81.52% and reaches an AUROC of 0.9022. The other models generally gain ACCs between 81% and 82%. Their MCC values fall within the range of 0.62 to 0.66, and their AUROC scores range from 0.8690 to 0.9004. These results show that the dual-path feature extraction and contrastive learning in SNN-ac4C enhance its capability to represent both local and global sequence features. This leads to a more balanced classification of true positives, true negatives, false positives, and false negatives. As a result, SNN-ac4C achieves higher overall ACC and maintains a better balance between SEN and SPE.

4.2 Ablation Studies

To further assess the effect of individual components in the SNN-ac4C model on ac4C site identification, we conduct ablation studies. The experiments systematically exclude core modules: BiLSTM, CNN, multi-head self-attention (MHSA), and contrastive learning (CL). As a result, four comparison models of SNN-ac4C are established: **No-BiLSTM**, **No-CNN**, **No-MHSA**, and **No-CL**.

Table 3 reports the ablation outcomes on validation dataset. Among all variants, No-BiLSTM exhibits the most noticeable performance drop, where SEN falls from 83.26% to 73.30% and MCC decreases from 0.6561 to 0.5927. These highlight the role of BiLSTM in capturing global contextual cues within mRNA sequences. The No-CNN model leads to a slight reduction in ACC (from 82.81% to 82.13%) and SPE (from 82.35% to 77.83%). It indicates that local feature extraction contributes to distinguishing modified from unmodified sequences. In the case of No-MHSA, SEN decreases from

83.26% to 81.90%, and MCC drops from 0.6561 to 0.6335, which suggests that MHSA helps model long-range dependencies.

Table 3. Prediction results on validation set for the ablation study.

Model	SEN(%)	SPE(%)	ACC(%)	MCC	AUROC
No-MHSA	81.90	81.44	81.67	0.6335	0.8904
No-CNN	**86.43**	77.83	82.13	0.6449	0.8884
No-BiLSTM	73.30	85.52	79.41	0.5927	0.8644
No-CL	79.19	**85.97**	82.57	0.6531	0.8854
SNN-ac4C	83.26	82.35	**82.81**	**0.6561**	**0.8911**

Table 4. Prediction results on test set for the ablation study.

Model	SEN(%)	SPE(%)	ACC(%)	MCC	AUROC
No-MHSA	86.59	80.98	83.79	0.6768	0.9010
No-CNN	86.41	78.62	82.52	0.6523	0.8885
No-BiLSTM	77.72	78.26	77.99	0.5598	0.8392
No-CL	85.69	80.62	83.15	0.6639	0.9008
SNN-ac4C	**87.68**	**81.52**	**84.60**	**0.6934**	**0.9022**

As for No-CL, the results show a slight decrease in ACC (from 82.81% to 82.57%) but an increase in SPE (from 82.35% to 85.97%). This non-uniform effect is explained by the function of Contrastive Learning (CL), which is to enhance feature discriminability and improve class separation. Without CL, the overall classification ability declines, as reflected in a lower ACC. However, the improved SPE suggests the model becomes more accurate in detecting negative samples. This shift may be due to the model adopting a more conservative decision boundary without CL to classify more samples as negative. While this raises SPE, it may also increase false negatives, leading to reduced ACC. This highlights the importance of CL in maintaining a balanced and precise classification performance.

Table 4 and Fig. 3 summarize the performance on the independent test set. No-BiLSTM causes SEN to drop from 87.68% to 77.72%, and MCC from 0.6934 to 0.5598. No-CNN results in a decrease in ACC (from 84.60% to 82.52%) and SPE (from 81.52% to 78.62%). For No-MHSA, SEN falls to 86.59% and MCC to 0.6768. The No-CL lowers ACC to 83.15% and MCC to 0.6639. These results are similar with those on the validation set.

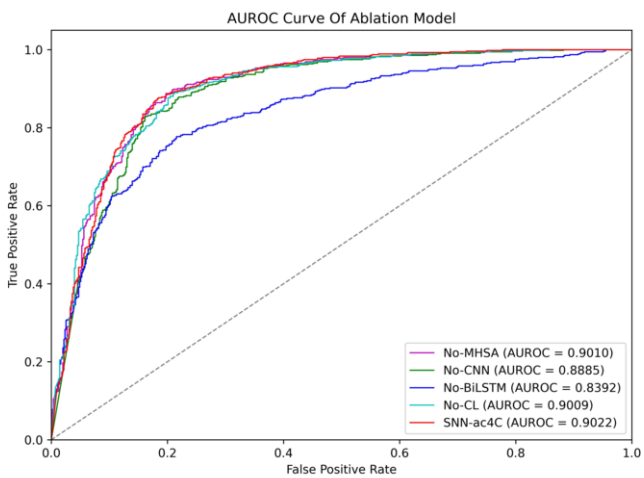


Figure 3. AUROC curves on test set for the ablation study.

The ablation experiments clearly demonstrate the contribution of each component within the SNN-ac4C framework. BiLSTM proves essential for modeling global sequence dependencies. Its absence notably impacts SEN and MCC. CNN serves as a local feature extractor, and its removal impairs both SPE and overall ACC. MHSA provides benefits in modeling complex sequence relationships. CL improves the separation between classes by enhancing feature embedding quality. In conclusion, all components in the SNN-ac4C architecture are necessary for optimal performance in ac4C site recognition. Results of all metrics drop when any module is removed.

4.3 *t-SNE-based Visualization*

We use t-SNE for visualization and examines how the SNN-ac4C model distinguishes between ac4C and Non-ac4C sites before and after training [11]. It applies t-SNE to the feature embeddings under two different conditions. The first is the *No-trained Output*, where the model has not been trained. The second is the *Trained Output*, where the SNN-ac4C is fully trained. In the *No-trained Output* (Fig. 4 A), ac4C (blue) and Non-ac4C (yellow) points distribute randomly across the plot. There is minimal clustering between the two classes. This result suggests that the initial feature space lacks discriminative ability. In contrast, the *Trained Output* (Fig. 4 B) displays two clearly separated clusters. These clusters correspond to the ac4C and Non-ac4C. The result indicates that the well-trained SNN-ac4C captures both local and global contextual features. This capability improves its ability to distinguish between modified and unmodified ac4C sites.

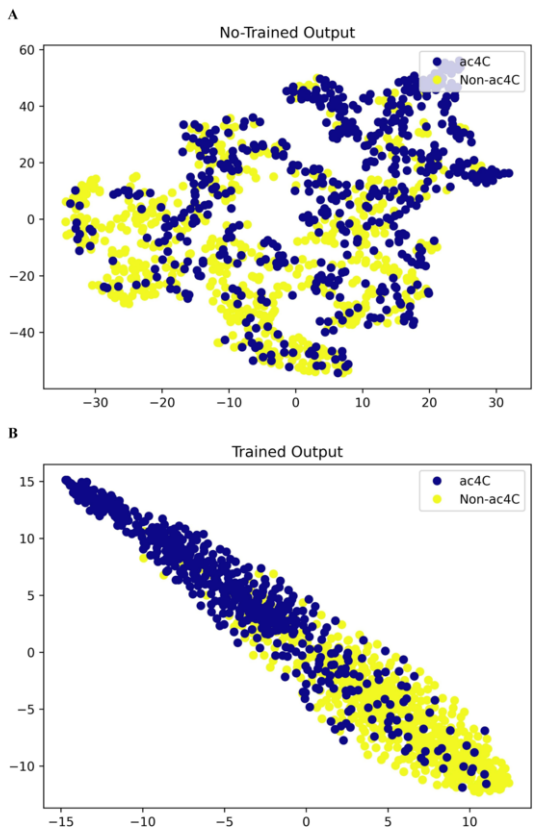


Figure 4. t-SNE results of SNN-ac4C.

4.4 *Comparison with Baselines*

Based on the results of the ablation studies and the t-SNE-based visualization, the effectiveness of SNN-ac4C is validated. We then evaluate its performance by comparing it with several existing ac4C

detection models. These baselines include Stacking-ac4C [24], LSA-ac4C [20], ac4C-AFL, DPNN-ac4C [39], and NBCR-ac4C. NBCR-ac4C is regarded as the current state-of-the-art (SOTA) approach. LSA-ac4C uses a dual-layer Long Short-Term Memory (LSTM) network combined with a self-attention module to enhance ac4C site recognition. Stacking-ac4C incorporates three feature encoding strategies (Kmer, PseKNC, and PseEIIP) and applies a stacking ensemble for classification. DPNN-ac4C adopts a self-attention structure for detecting ac4C sites in mRNA sequences.

Table 5 displays the performance of all models on the independent test set. SNN-ac4C achieves the highest classification ACC at 84.60% and obtains the best MCC of 0.6934. These scores surpass those of NBCR (83.51% ACC, 0.6706 MCC) and DPNN-ac4C (82.78% ACC, 0.6578 MCC), which highlights its strong overall prediction ability. In terms of true positive rate, SNN-ac4C records the highest SEN at 87.68%. This implies its effective detection of ac4C sites. Although DPNN-ac4C reaches a slightly higher AUROC of 0.9103, SNN-ac4C shows better balance across all metrics. It maintains an SPE of 81.52%, and the combination of its higher ACC and MCC reflects a more reliable result in real-world scenarios.

We acknowledge that SNN-ac4C has a slightly lower AUROC than DPNN-ac4C (0.9022 vs 0.9103). However, this trade-off should be analyzed in the context of practical biological applications. While AUROC measures the model's ability to distinguish between classes across different threshold settings, ACC and MCC provide more direct measures of classification performance at optimal operating points. In biological sequence analysis, ACC and MCC are often considered more robust metrics for practical applications, as they reflect a better balance between sensitivity and specificity that is crucial for real-world decision-making. The superior ACC (84.60% vs 82.78%) and MCC (0.6934 vs 0.6578) of SNN-ac4C compared to DPNN-ac4C demonstrate that our model achieves more reliable overall performance, particularly in scenarios where balanced classification is essential for biological interpretation and downstream analysis.

We clarify that the performance gain results from the integration of multiple complementary modules within SNN-ac4C. The BiLSTM layer captures global sequence dependencies and the convolutional layer extracts local features efficiently. The MHSA component learns complex interactions among sequence elements. Contrastive learning, we believe the most important module, sharpens feature distinction by aligning similar samples while pushing apart dissimilar ones in the representation space. These combined modules enable SNN-ac4C to learn robust and discriminative features, which contributes to its superior performance compared with baselines.

Table 5. Prediction results on the test set, compared with different methods.

Model	SEN(%)	SPE(%)	ACC(%)	MCC	AUROC
Stacking-ac4C	80.80	80.80	80.80	0.6159	0.8835
LSA-ac4C	87.13	78.26	82.70	0.6566	0.8953
ac4C-AFL	84.40	80.30	82.30	0.6470	0.8950
DPNN-ac4C	81.78	**84.78**	82.78	0.6578	**0.9103**
NBCR-ac4C	84.96	82.07	83.51	0.6706	0.8958
SNN-ac4C	**87.68**	81.52	**84.60**	**0.6934**	0.9022

4.5 Robustness of SNN-ac4C

We further assess the robustness of SNN-ac4C by conducting additional experiments on an independent dataset derived from PACES [42]. This dataset contains experimentally validated ac4C sites in human mRNA. Unlike the 201-nucleotide sequences used in previous experiments, this dataset consists of sequences with 415 nucleotides that introduce greater complexity. We apply CD-HIT with a threshold of 0.4 to eliminate highly similar sequences and reduce the risk of overfitting. The final filtered dataset from PACES has 1,148 positive and 5,439 negative samples for training. The independent test set contains 467 positive and 2,151 negative samples. We also establish 10 balanced subsets that each contains an equal number of positive and negative samples in both training and testing sets.

In comparison, several deep learning-based models are selected to predict ac4C site prediction in this independent dataset from PACES. DeepAc4C sets convolutional neural network (CNNs) and builds hybrid features to combine physicochemical patterns with distributed nucleotide representations [32]. EMDL-ac4C adopts an ensemble framework based on a modified DenseNet. It also fuses convolutional residual blocks and Squeeze-and-Excitation networks [15].

Table 6 presents the performance comparison. SNN-ac4C achieves the best performance among all models. It reaches 81.43% in ACC and 0.6301 in MCC. In comparison, DeepAc4C reports 79.19% in ACC and 0.5857 in MCC. TransC-ac4C obtains 80.69% and 0.6146, while EMDL-ac4C yields 80.80% and 0.6169, respectively. Although all models perform similarly in terms of AUROC, the higher ACC and MCC of SNN-ac4C suggest a better balance between recall and specificity that supports dependable predictions.

We argue the consistent robustness of SNN-ac4C across datasets comes from its architecture including BiLSTM for modeling global dependencies, CNN for extracting local patterns, self-attention for capturing long-range interactions, and contrastive learning for enhancing feature discrimination. These modules enable it to adapt effectively to the input dataset. The stable performance across all metrics reveals the reliability of SNN-ac4C for identifying ac4C sites, even when sequence lengths differ from previously tested conditions.

Table 6. Prediction results on the independent dataset compared with current models.

Model	SEN(%)	SPE(%)	ACC(%)	MCC	AUROC
DeepAc4C	82.80	75.58	79.19	0.5857	0.8649
TransC-ac4C	80.94	80.45	80.69	0.6146	0.8691
EMDL-ac4C	81.04	**81.73**	80.80	0.6169	**0.8794**
SNN-ac4C	**82.85**	80.02	**81.43**	**0.6301**	0.8784

5 Conclusion

This study introduces SNN-ac4C, a dual-path deep learning framework for identifying ac4C sites in mRNA. It integrates BiLSTM, CNN, multi-head self-attention, and contrastive learning to model both local sequence patterns and global dependencies. Experiments on the test set show that SNN-ac4C achieves state-of-the-art performance, with 84.60% accuracy and 0.6934 MCC. Ablation studies validate the contribution of each module, while visualization reveals clear feature clustering between ac4C and Non-ac4C sites. SNN-ac4C also generalizes well to external datasets with varying sequence lengths. Future work includes expanding to other modifications and adding structure-based inputs.

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